ETALE SECTIONS OVER THE BASE AGREE WITH SECTIONS OVER A FLAT QUASI-COMPACT COVER WITH CONNECTED GEOMETRIC FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the injection, the fpqc reduction and the geometric-fibre computation are all inside the counted span, and the author's closing phrase is retained verbatim.

Editorial note. The label lemma-sections-upstairs occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Small inverse image f_{small}^{-1} of a sheaf on the small etale site. Same chapter, the pullback f_{small}^{-1} of a sheaf on S_etale along a morphism f : T -> S. Supplies the sheaf whose sections over T are compared with the sections over S.
- fpqc coverings and the description of the pullback of a section. Same chapter, the fact that a flat quasi-compact surjective morphism gives an fpqc covering and the resulting description of pullbacks of sections. Supplies the reduction of surjectivity to agreement of the two pullbacks to T x_S T.
- Constant sheaf on a connected scheme and constancy of its sections. Same chapter, the constant sheaf with value F_sbar on the small etale site of a connected scheme, whose sections are constant. Supplies the identification of the two pullbacks over each geometric fibre, which is the crux of the proof.

Source reference. Original lemma 09XN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4960–4960. Same-chapter navigation pointer invoked at proof lines 5049-5050, where the author uses the description of the pullback of a section along an fpqc covering. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let S be a scheme. Let $f : T \to S$ be a morphism such that*

(1) *f is flat and quasi-compact, and*

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(2) *the geometric fibres of f are connected.*

Let $\mathcal{F}$ be a sheaf on $S_{étale}$. Then $\Gamma(S, \mathcal{F}) = \Gamma(T, f_{small}^{-1}\mathcal{F})$.

Proof. There is a canonical map $\Gamma(S, \mathcal{F}) \to \Gamma(T, f_{small}^{-1}\mathcal{F})$. Since f is surjective (because its fibres are connected) we see that this map is injective.

To show that the map is surjective, let $\alpha \in \Gamma(T, f_{small}^{-1}\mathcal{F})$. Since $\{T \to S\}$ is an fpqc covering we can use Lemma 09XN to see that suffices to prove that α pulls back to the same section over $T \times_S T$ by the two projections. Let $\overline{s} \to S$ be a geometric point. It suffices to show the agreement holds over $(T \times_S T)_{\overline{s}}$ as every geometric point of $T \times_S T$ is contained in one of these geometric fibres. In other words, we are trying to show that $\alpha|_{T_{\overline{s}}}$ pulls back to the same section over

$$(T \times_S T)_{\overline{s}} = T_{\overline{s}} \times_{\overline{s}} T_{\overline{s}}$$

by the two projections to $T_{\overline{s}}$. However, since $\mathcal{F}|_{T_{\overline{s}}}$ is the pullback of $\mathcal{F}|_{\overline{s}}$ it is a constant sheaf with value $\mathcal{F}_{\overline{s}}$. Since $T_{\overline{s}}$ is connected by assumption, any section of a constant sheaf is constant. Hence $\alpha|_{T_{\overline{s}}}$ corresponds to an element of $\mathcal{F}_{\overline{s}}$. Thus the two pullbacks to $(T \times_S T)_{\overline{s}}$ both correspond to this same element and we conclude. □